Speciation Sandbox

Finding out what actually decides whether a separated population becomes a separate species — and why "almost isolated" is not almost there.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/speciation-sandbox/

1 Before you open anything

Answer from what you already know. Having a specific claim on paper is what makes the rest of the sheet worth doing.

PREDICTION

A mountain range splits one population of 100 animals into two populations of 50. They stay apart long enough that **96% of the gene flow between them is now blocked**: most pairings are refused, and the hybrids that do get born mostly fail to survive or breed. The mountain range then erodes away, permanently, and the two populations meet again.

Are they now two species that stay two, or do they merge back into one population? Commit to one.

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Write one or two sentences saying what reasoning got you there.

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2 One split, start to finish

Open the demonstration. Two populations of 50 sit either side of a corridor. The panel *Where the two populations stand* reports four numbers about them; the panel *If the barrier came down right now* reports one more.

1. Press **Reset to one population**. Leave both sliders where they are — **Gene flow through a raised barrier** at **0.00** and **Differing selection** at **3.0×10^{-3}**. Record the top row of the table.
2. Set the speed control to **Slow**.
3. Press **Raise the barrier**. The clock starts on its own.
4. Press **Pause** as close to **60** as you can get on the **Generations separated** readout. Write down the number you actually stopped at, then record the row.

5. Press **Lower the barrier**. This restarts the clock as well.
6. Press **Pause** once **Generations in contact** passes about **40**, and record the last row.

When	**Genetic divergence**	**Reproductive isolation index**	**Hybrid fitness**	**Gene flow actually getting through**
Barrier down, generation 0				
Barrier up, about 60 generations apart (actual: ________)				
Barrier down again, about 40 generations of contact				

a. At generation 0 the barrier is down and divergence is not zero. Read the caption under the map and write the value it settles at, together with the reason divergence does not fall all the way to zero.

b. With the barrier up, *Gene flow actually getting through* reads 0.00, yet divergence keeps climbing. Name the two processes listed in the *Why they diverge at all* panel that are driving it, and state which of the two would still operate if both habitats were identical.

c. While you were paused at 60 generations apart, the panel *If the barrier came down right now* gave a number and a sign. Write both, and say in one sentence what the sign was telling you would happen.

3 How long is long enough?

Now change one thing only: how many generations the barrier stays up before it comes down. Everything else stays where it was in Part 2.

1. Press **Reset to one population**, keep the speed on **Slow**, and press **Raise the barrier**.
2. Press **Pause** near the target generation count in the first column. Record divergence, the isolation index, and the number and sign shown under *Divergence would change by*.
3. Press **Lower the barrier** and let it run for at least 60 generations of contact. In the last column write *fused* if divergence falls back near the generation-0 value, or *stayed two* if it holds high or climbs.
4. Press **Reset to one population** and repeat for the next row.

Generations apart	Genetic divergence	Reproductive isolation index	Divergence would change by (with sign)	Outcome after contact
60				
110				
140				
200				

a. In which row does the sign of *Divergence would change by* first stop being negative? Give the isolation index at that row to three decimal places.

b. State the rule in your own words: what has to be true at the moment the barrier comes down for the split to survive contact? Do not answer with a number of generations.

c. Look at your 110-generation row. Its isolation index is close to 0.96, which is the situation you predicted in Part 1. What actually happened, and if it differs from your prediction, name the specific assumption that turned out to be incomplete.

> Run the 140-generation row a second time from a fresh **Reset to one population**. The outcome can differ between runs at that separation, because the model adds a small random wobble to drift. Two runs disagreeing is a result, not a mistake — record both.

4 A barrier that leaks

The *A leaky barrier* panel sets how many migrants per generation cross even while the barrier is up. Its readout, *Divergence this leak settles at*, is a closed form you can compute on paper: r / (r + 2Nm/N), where r is the divergence pressure and N = 50.

1. Drag **Differing selection** all the way left, to **0.0×10^{-3}**. With selection off, r = 1/(2N) = 0.010 and the readout reduces to **1 / (1 + 4Nm)**.
2. Before touching the leak slider, compute all five predicted values on paper.
3. Now set **Gene flow through a raised barrier** to each value in turn and record what the readout says.

Leak (Nm, migrants per generation)	Your prediction from 1/(1+4Nm)	Readout
0.25		
0.50		
1.00		
2.00		
4.00		

a. Working backwards: solve 1/(1+4Nm) = 0.100 for Nm on paper, then set the slider to your answer and confirm the readout. Show the algebra.

A common shortcut says a barrier that blocks most movement is nearly as good as a complete one. With the barrier down, 10 migrants cross per generation, so a leak of 1.00 is a barrier blocking 90% of movement. Test that shortcut.

1. Put **Differing selection** back to **3.0×10^{-3}** and the speed control on **Fast**.
2. Set the leak to **1.00**, press **Reset to one population**, then **Raise the barrier**, and let it run past **400** generations separated. Record the row.
3. Repeat with the leak set to **0.25** — a barrier blocking 97.5%.

Leak	Movement blocked	Genetic divergence at ~400 generations	Reproductive isolation index
1.00	90%		
0.25	97.5%		

b. One of those two runs stops changing and stays put no matter how long you wait. Which one, and why does waiting longer not help it?

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c. A mountain range stops 90% of the movement between two populations. Using your table, explain in one or two sentences why calling it "almost a complete barrier" gives the wrong answer about what will happen.

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5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself — and where you were wrong, write down which step went off.

You may use: each population has $N = 50$; with the barrier down 10 migrants cross per generation; drift alone contributes $1/(2N)$ per generation; with drift only, divergence settles at $1/(1 + 4Nm)$; and from Part 3, contact stops undoing a split only once divergence is past roughly 0.84, which is an isolation index of about 0.994.

1. Two populations of 50 lizards are separated by a canyon that still lets 2 individuals cross per generation. Selection in the two canyon walls is indistinguishable, so treat divergence as drift-driven only. Calculate the divergence they settle at, compare it to the 0.84 turning point, and state whether they will ever become separate species while the canyon stays as it is.

2. A reserve manager wants two fragmented populations of 50 frogs each to stay one species, and sets a target of holding divergence at or below 0.20. Assuming drift only, calculate the minimum number of migrants per generation that has to be moved between the fragments. Express your answer both as migrants per generation and as a percentage of one population per generation.

3. A field guide describes two bird populations on either side of a gorge as "96% reproductively isolated — almost a separate species." Using your Part 3 data, explain why that phrasing predicts the wrong outcome. Include the number of migrants per generation that still land their genes at 96%, and state roughly what the isolation index would have to reach before the split survives the gorge closing. No calculation is required beyond that one multiplication.

4. A classmate writes: "Once the barrier went up, the two populations changed so that they would no longer be able to breed with each other." Rewrite that sentence so it describes what the model actually does, and then explain in one sentence why the original wording misstates how mutation and selection are related in time.

> The reproductive isolation index is invented for this page. No real study reports a single number that ticks from 0 to 1; the separate things measured are whether the two forms will court, whether fertilisation happens, whether hybrids survive, and whether hybrids are fertile. Everything here is also **allopatric** speciation only. Speciation happens without any physical barrier as well — polyploidy in plants, host-race shifts in insects, assortative mating within one lake — and nothing on this page models that.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Press **Split 140 gen** five times in a row, resetting between runs, and keep a tally of how many times the populations end up two. Then decide what that tally does to the idea that a species has a founding date.
- On the History plot, the isolation curve starts below the divergence curve and later crosses above it. Find the divergence value where they cross, and connect the shape to hybrid fitness falling as $\exp(-8D^5)$ rather than in proportion to D.
- Push **Differing selection** to its maximum, **10.0×10^{-3}**, and rerun a 60-generation split. Compare how far it gets against your Part 2 row, and work out how much of that difference comes from selection rather than drift.
- Watch the corridor on the map during contact after a long split. The hybrid circles get drawn crossed out below a certain hybrid fitness. Find roughly what value that happens at.